A Novel Spectral Index for Mapping Sunflowers Using Sentinel-2 Imagery

Qingji Meng, Shuying Zang, Bingxue Zhu, Lushi Liu, and Kaishan Song

***Abstract*—As a globally significant oilseed and economic crop, accurately mapping the spatial distribution of sunflowers is crucial for ensuring food security. However, existing sunflower classification models are often complex in design, and specialized vegetation indices (VIs) remain relatively scarce. Therefore, there is an urgent need to develop simple, efficient, and precise identification methods. This study introduces the sunflower identification index (SII), developed using the RE2 band, which is sensitive to chlorophyll variation, and the shortwave infrared 2 (SWIR2) band, which reflects differences in canopy water content. The SII provides critical spectral information necessary for distinguishing sunflowers from other crops. The effectiveness of the SII was evaluated in three major sunflower-producing countries—China, Ukraine, and France—across three representative sunflower-growing regions, each characterized by distinct climatic conditions and farming systems. The results indicate that in the Hetao Irrigation District (S1) in 2023, the classification accuracy of SII (overall accuracy (OA) = 94.39% and Kappa coefficient (Kappa) = 0.89) was significantly higher than that of the normalized difference vegetation index (NDVI) and the green NDVI (GNDVI). Furthermore, the SII continued to exhibit excellent performance when applied across different years and regions (the Ukrainian site (S2) and Toulouse (S3) in 2022, S1 in 2021, and S3 in 2020). Finally, the SII was employed in binary classification using both the random forest (RF) and support vector machine (SVM) algorithms, yielding satisfactory results (OA > 90% and Kappa ≥ 0.81). These findings demonstrate that the SII possesses strong spatiotemporal transferability capabilities and robustness, allowing it to maintain high stability in complex and dynamic agricultural environments. In the future, integrating the SII with deep learning (DL) algorithms is expected to further enhance classification accuracy.**

***Index Terms*—Identification, index, machine learning (ML), remote sensing, sunflower.**

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I. Introduction

As a globally significant oilseed and economic crop [1], the sunflower is widely cultivated in 74 countries, with a total planting area of approximately 29 million hectares and an annual production of 54 million tons (FAO 2022). Sunflowers serve not only as a crucial source of edible oil for humans and a key raw material for industrial production [2], [3] but also, due to their unique ornamental value, as an important tourism resource, playing a significant role in promoting the development of local agricultural economies [4]. However, influenced by factors such as climate change, the spatial distribution of sunflower cultivation has undergone significant changes in recent years, while global demand for sunflowers and their derived products continues to rise [5], [6]. Against this backdrop, the timely and accurate monitoring and mapping of sunflower cultivation are of great importance for ensuring food security, stabilizing market supply and demand, and formulating scientific agricultural policies.

In comparison with conventional field surveys for acquiring crop information, remote sensing technology has been extensively employed in crop identification studies due to its merits of extensive coverage, minimal expense, and prompt data updates [7], [8]. When selecting data sources, it is essential to comprehensively consider factors such as research scale, data availability, and crop phenological characteristics. Advancements in remote sensing technology have led to the launch of various medium- to high-resolution satellites, including MODIS [9], Landsat [10], [11], and Sentinel-2 [12], [13]. Among these, Sentinel-2 has emerged as the preferred choice for crop classification mapping across a range of research scales due to its exceptional performance [14], [15], [16]. Its high spatial resolution and rich spectral bands enable precise crop identification in mountainous regions and fragmented farmland, while also providing more classification features for distinguishing between different crops [17], [18]. Additionally, Sentinel-2's short revisit cycle facilitates the generation of a more comprehensive time-series dataset, which is instrumental for crop classification. These data not only capture the dynamic changes in crop growth but also reveal subtle differences among crops under varying seasonal and environmental conditions, thereby offering more detailed and comprehensive classification information [19], [20], [21]. However, cloud contamination remains an important factor affecting the stable acquisition of optical remote sensing imagery. Therefore, in crop classification studies, image selection strategies are typically optimized, and effective observations during key

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crop phenological stages are exploited to further improve classification accuracy.

In recent years, as remote sensing research has advanced toward greater precision, intelligence, and quantification, traditional methods have gradually become insufficient to meet growing demands. To address this challenge, machine learning (ML) and deep learning (DL) have been widely applied to enhance classification accuracy and efficiency. In the domain of crop classification, ML has been demonstrated to be effective in leveraging medium- to high-resolution imagery for high-precision crop identification [22], [23], [24]. Decision tree (DT) is a classic ML algorithm that has been shown to facilitate rapid and precise identification of target crops in complex land cover environments by integrating multiple classification features, thereby ensuring a clear and interpretable classification process [25], [26], [27]. Advancements in computer technology have led to the widespread use of ML algorithms such as random forest (RF) and support vector machine (SVM) for crop classification based on remote sensing data [19], [28], [29]. These algorithms have been shown to efficiently process multidimensional data and exhibit strong generalization capabilities, making them valuable tools for improving classification accuracy and efficiency [30], [31]. However, ML algorithms are not infallible and are subject to limitations in their ability to extract deep features from imagery, which can pose challenges in complex scenarios. In contrast, DL, through automatic feature extraction, can capture more detailed and intricate features from data, demonstrating significant potential, particularly in crop identification [32], [33], [34]. Nevertheless, the implementation of DL algorithms necessitates substantial computational resources and large amounts of labeled data, which hinders their widespread application in large-scale studies [35], [36].

Vegetation indices (VIs) are instrumental in the extraction of vegetation information from remote sensing data. Due to their high sensitivity to vegetation growth variations and strong classification performance, they have been extensively utilized in crop classification studies [15], [37], [38]. Specifically, the normalized difference vegetation index (NDVI) and green NDVI (GNDVI) have exhibited remarkable efficacy in sunflower classification. For instance, using NDVI time-series data derived from HJ-1A/1B imagery, an asymmetric logistic curve was applied to model NDVI variations in crops, and phenological metrics were extracted. A classification ellipse model was then employed to achieve high-precision sunflower identification in the Hetao Irrigation District [39]. Additionally, the Skcnn_Tabnet model, which is based on NDVI and multiband features extracted from single-date Sentinel-2 imagery, accurately classified crops such as sunflowers and maize in Xing'an League and the Hetao Plain [29]. Likewise, leveraging Sentinel-2 time-series data and VIs such as NDVI and GNDVI, in conjunction with DL models, enabled early and highly accurate sunflower classification in the Shiyang River Basin by late July [40]. Furthermore, studies have demonstrated the efficacy of the b6–b8a bands of Sentinel-2 in sunflower identification [32], and VIs incorporating red-edge bands have been shown to outperform those relying solely on red bands [41]. Another study, employing Sentinel-1 multitemporal data, constructed a sunflower classification model by selecting key classification features and integrating them with an RF algorithm. This model has demonstrated the capability to achieve high-precision identification across diverse time periods and geographical locations [21]. In summary, the extant studies have contributed significantly to sunflower classification. Nevertheless, there is considerable potential for further exploration and enhancement in this domain. First, VIs such as NDVI and GNDVI are typically used as just one of many classification features for sunflower identification, while few studies have focused on developing VIs specifically tailored for sunflower recognition based on the differential sensitivity of spectral bands to vegetation physiological characteristics. Second, existing classification methods frequently encounter challenges in offering classification time windows that are widely applicable. These methods typically require substantial datasets and a multitude of classification features, leading to intricate model designs and substantial computational costs, which hinder their broad application. Finally, under the same classification feature conditions, most studies focus on comparing the strengths and weaknesses of different classification methods, while fewer have assessed the robustness of the classification features themselves.

In light of the aforementioned points, this study develops the sunflower identification index (SII) based on differences in chlorophyll content (RE2) and canopy water content shortwave infrared 2 (SWIR2) between sunflowers and other crops. This index offers a stable and widely applicable classification time window, while concurrently achieving a substantial reduction in data requirements. Furthermore, an evaluation was conducted to ascertain the accuracy, stability, and robustness of the SII in identifying sunflowers across three major sunflower-producing countries. The main contents of this study are as follows.

1) The SII was developed utilizing the RE2 and SWIR2 bands. Its performance in sunflower identification was evaluated by comparing it with NDVI and GNDVI in the Hetao Irrigation District (S1) in 2023, based on the DT algorithm.
2) The SII was applied to sunflower identification in S2 and S3 for 2022, S1 for 2021, and S3 for 2020 to evaluate its spatiotemporal transferability, with percentage error used to assess the accuracy of the classified area. Meanwhile, SII was combined with RF and SVM to perform binary classification across S1–S3, in order to further evaluate its robustness.
3) The advantages of SII were explored from multiple perspectives, including its construction principles, the benefits of its classification time window, and its adaptability and stability across different times and spatial scales. Meanwhile, its limitations were analyzed, and potential future applications were discussed.

II. Materials

A. Study Areas

The study area for index development, the Hetao Irrigation District (S1), is located in Bayannur, Inner Mongolia

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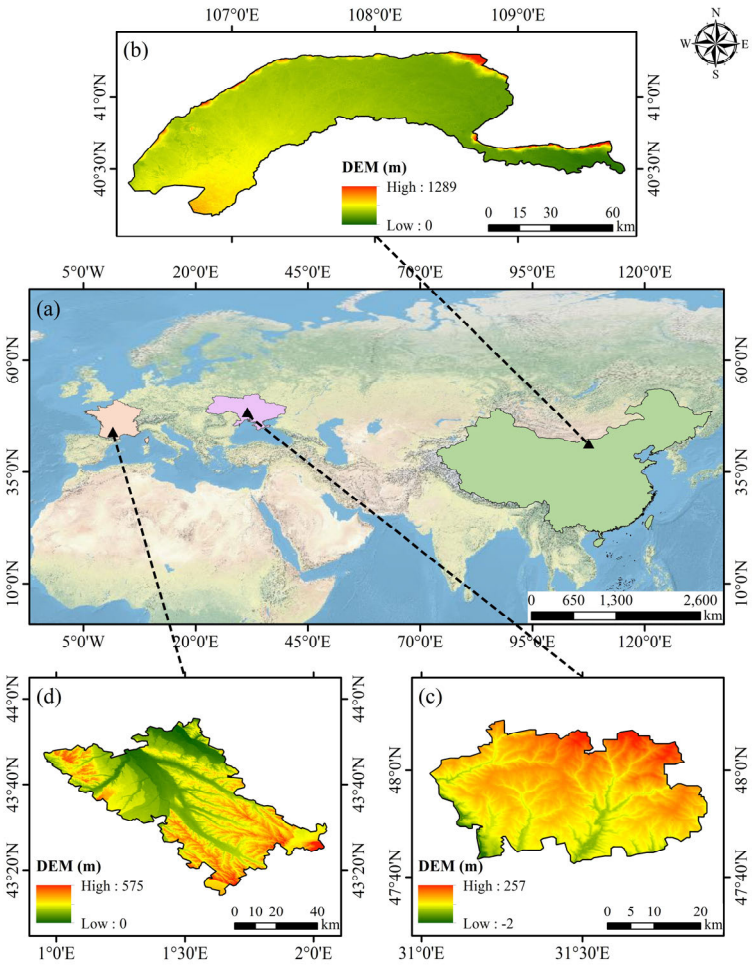


Fig. 1. Location map of the study areas. (a) Geographic locations of the three study areas (indicated by black triangles). (b) Hetao Irrigation District, China (S1). (c) Arbuzynka and Bratske, Ukraine (S2). (d) Toulouse, France (S3). Background image: World Physical Map.

TABLE I

BASIC INFORMATION ON THE STUDY AREAS AND THE NUMBER OF SATELLITE IMAGES USED

Study sites	Location	Area covered (km^2)	Number of images	Cropping system	Year
S1	Inner Mongolia Autonomous Region, China	14047	16	Single	2023
			2		2021
S2	Mykolaiv oblast, Ukraine	2549	18	Crop rotation	2022
S3	Haute-Garonne, France	2047	15	Crop rotation	2022
			2		2020

Autonomous Region, China (see Fig. 1(a) and (b) and Table I). The Hetao Irrigation District is the third-largest irrigation district in China, with a total irrigated area of 570 000 hectares. It is one of the country's most significant sunflower cultivation regions. The region's climate is typified by a temperate continental climate, with an average annual precipitation of approximately 160 mm [39], while evaporation reaches as high as 2100–2300 mm. The mean annual temperature ranges from 6 °C to 10 °C [42]. Statistical data and related studies indicate that the primary crops in this region are maize, sunflower, spring wheat, and zucchini, with sunflower intercropping being a common practice [32], [39], [43].

The Ukrainian site (S2) consists of two counties in Mykolaiv Oblast: Arbuzynka and Bratske [Fig. 1(c)]. This region is a prominent grain production base in Ukraine, characterized by a temperate continental climate with relatively low annual precipitation. The mean summer and winter temperatures are approximately 24 °C and −2 °C, respectively. Sunflower cultivation occupies the most extensive area in this region, followed by winter wheat, maize, and winter rapeseed (EUCROPMAP 2022).

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Toulouse (S3) is located in the northeastern part of Haute-Garonne, France [Fig. 1(d)], in the transition zone between a Temperate Oceanic climate and a Mediterranean climate. The region is characterized by a flat and open landscape, which offers favorable hydrothermal and topographical conditions for agricultural cultivation. Among various summer crops, sunflower is the most extensively planted, covering 31% of the agricultural land, followed by maize (11%) and soybean (3%) (EUCROPMAP 2022). The cropping structure in these regions is characterized by complexity, with crop rotation being widely practiced and sunflowers frequently included.

B. Data

1) Acquisition and Preprocessing of Sentinel-2 Imagery: Sentinel-2 is an optical remote sensing satellite equipped with a multispectral instrument (MSI) and comprises three satellites (2A, 2B, and 2C). It provides 13 spectral bands covering the visible, near-infrared (NIR), and SWIR regions, with varying spatial resolutions (10, 20, and 60 m). Of particular significance are the three red-edge bands, which play a crucial role in vegetation monitoring. However, the coastal aerosol band (B1), the water vapor band (B9), and the cirrus cloud band (B10) have limited utility for vegetation analysis. Therefore, a total of ten bands were selected for crop classification.

Due to incomplete coverage issues with Sentinel-2 Level-2A data from European Space Agency (ESA) Copernicus Data Space (https://browser.dataspace.coperni-cus.eu/). Therefore, Level-1C data were selected for crop classification. The selection of images for each study area was determined based on the growing periods of sunflower and other major crops. For S1, 16 available images from May to September 2023 were used to develop and validate the classification performance of the SII. In addition, to evaluate its temporal transferability across different years within the same study area (i.e., the annual stability of thresholds) and considering image availability, two further images from August 19 and August 29, 2021, were selected (Table I). For S2, 18 images from March to September 2022 were selected. Since, apart from EUCROPMAP 2022, no other multiyear, high-quality, open-access, and free crop datasets or county-level statistics covering Ukraine are available for validation [44], cross-year validation based on threshold consistency was not conducted in this region. For S3, 15 images from March to October 2022 were selected, along with two additional images from July 9 and August 8, 2020, for cross-year validation of threshold consistency.

During the data preprocessing stage, the Sen2Cor-2.11.0 tool was used to perform atmospheric correction on all acquired Level-1C images, generating Level-2A products. Subsequently, the Level-2A images were resampled to a spatial resolution of 10 m using SNAP 9.0. Given the necessity of acquiring full coverage of each study area, the resampled data were mosaicked and clipped sequentially using ENVI 5.6 to obtain complete and usable imagery.

2) Sample Point Data: From September 6 to 11, 2023, a field survey was conducted on crop planting structures in S1. A handheld GPS device was utilized for positioning, as detailed in Table II. To ensure the reliability of the sample points, all points have a positioning accuracy error of less than 5 m. Given the extensive cultivation area and widespread distribution of the crop, 111 and 178 sample points were collected for maize and sunflower, respectively. In contrast, zucchini, distinguished by its comparatively limited planting area and dispersed distribution, yielded a mere 47 sample points. During the survey period, the spring wheat had already been harvested, necessitating its identification based on residual surface remains, which, to some extent, limited the number of sample points collected for this crop.

With the exception of a subset of sample points in S3, which were derived from the LUCAS 2022 field survey data (https://ec.europa.eu/eurostat/web/lucas/databa-se/2022), all sample points in S2 and the majority in S3 were obtained through manual annotation of EUCROPMAP 2022 data (Table II). The crop types investigated in this study correspond to the major crops included in EUCROPMAP 2022, and the cropping structures in S2 and S3 exhibit a high degree of similarity. Furthermore, the sample points for each crop type were randomly partitioned in a 7:3 ratio, with 70% designated as training data and the remaining 30% used as validation data to assess classification accuracy. It should be noted that when evaluating the temporal transferability of the SII (i.e., S1 in 2021 and S3 in 2020), an independent accuracy assessment of the cross-year results could not be performed due to the lack of available validation samples. Therefore, this study employed alternative validation approaches, including enlarged visualization of representative plots, consistency comparison with reference data, and accuracy assessment of the classified area, to evaluate the reliability of the cross-year transfer results. Relevant details are provided in Sections IV-B and IV-C.

3) Land Use Data: The ESA WorldCover 2021 product provides a global land cover map for 2021 at a 10-m resolution, generated using Sentinel-1 and Sentinel-2 data. It has an overall accuracy (OA) of 77% (https://doi.org/10.5281/zenodo.7254221) [45]. All study areas exhibit intricate land use conditions, encompassing not only diverse crop types but also a variety of natural and artificial features. Consequently, the cropland data from ESA WorldCover 2021 were employed to mask the preprocessed Sentinel-2 imagery, thereby effectively reducing the influence of nonagricultural land cover on the accuracy of sunflower classification.

EUCROPMAP 2022 is a 10-m resolution crop type map developed by European Commission's Joint Research Centre (JRC). It provides information on 19 crop types and six noncrop classes across the 27 EU countries and Ukraine (http://data.europa.eu/89h/555e5d1d-1aae-4320-a716-2e6d18aa1e7c). This dataset is generated from Sentinel-1 and Sentinel-2 imageries, combined with ground survey data and the RF algorithm, aiming to support agricultural monitoring, policy-making, and land use analysis across Europe [46]. In this study, the dataset was used not only to obtain sample points but also as reference data for comparing the sunflower classification results from S2 to S3 (2022) and for verifying the reliability of the classified areas (see Sections IV-B and IV-C).

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TABLE II
CROP TYPES AND THE NUMBER OF SAMPLE POINTS IN EACH STUDY AREA

S1		S2		S3	
Crop type	Quantity	Crop type	Quantity	Crop type	Quantity
Spring wheat	10	Winter wheat	68	Winter wheat	109
Maize	111	Maize	35	Maize	31
Sunflower	178	Sunflower	89	Sunflower	70
Zucchini	47	Winter rapeseed	20	Winter rapeseed	20
		Soybean	8	Soybean	15
		Other crops	26	Other crops	32
Total	346		246		277

The Theia Land Cover Map 2020, produced by the Theia Land Data Centre in France, is an annual land cover dataset with a spatial resolution of 10 m (https://zenodo.org/records/6538861). This dataset is derived from Sentinel-2 imagery and generated using time-series analysis combined with the RF classification approach, providing consistent and continuous nationwide land cover information for France. The dataset comprises 23 land cover classes, among which the crop categories mainly include sunflower, maize, soybean, cereals, and rapeseed. In this study, the dataset was employed as reference data to verify the accuracy of the sunflower classified area for 2020 obtained through cross-year transfer using the thresholds determined from S3 in 2022.

4) Statistical Data: The statistical data used in this study were obtained from Bayannur Municipal Bureau of Statistics (http://tjj.bynr.gov.cn/) and were employed to verify the reliability of the sunflower classification results for 2021 and 2023 in the Hetao Irrigation District (S1). Although S1 covers multiple counties, only Wuyuan County, Linhe District, and Hangjinhou Banner were fully included within the study area; therefore, the area validation was primarily conducted for these three counties.

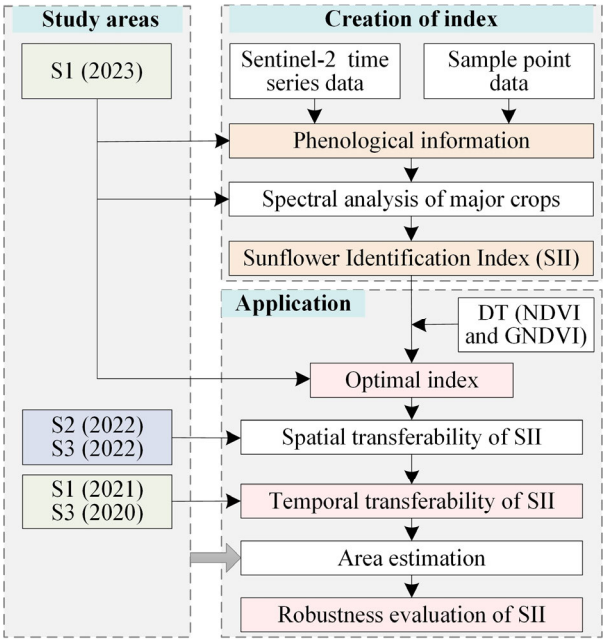


Fig. 2. Workflow of sunflower mapping.

III. METHODS

In this study, an SII was developed based on the RE2 (B6) and SWIR2 (B12) bands of Sentinel-2 data. This index enables accurate sunflower identification using imagery from the late squaring to early filling growth stages. The study's workflow comprises three primary components (Fig. 2).

1) *Development Process of the SII:* The phenological characteristics of sunflower and other crops were analyzed, the optimal time window and sensitive bands for sunflower identification were determined, and the spectral differences of various crops in the B6 and B12 bands were compared.
2) Using the DT algorithm as a baseline and combining it with the optimal threshold, the optimal index was selected in S1 (2023), and its spatiotemporal transferability as well as the accuracy of the classified area were evaluated.
3) Combination of the SII with RF and SVM classifiers for classification in S1–S3 to evaluate its robustness across different classifiers. It is imperative to acknowledge that the ensuing analyses and discussions are predicated on the Hetao Irrigation District (S1).

A. Phenological Information of Crops

In S1, sunflowers are sown in late May and subsequently progress through the phenological stages of emergence, seedling, squaring, flowering, filling, and senescence (Fig. 3). The growth period of spring wheat spans from April to July, while maize and zucchini grow from May to September. According to the phenological data provided by IPAD (https://ipad.fas.usda.gov/ogama-ps/cropcalendar.aspx), the growth cycle of sunflowers in S2 and S3 lasts from April to September, with each stage occurring approximately one month earlier than in S1 (Fig. 4). The growth period of maize extends from April to October, while winter wheat grows from October to the following July, winter rapeseed from October to the following June, and soybean from April to October. As illustrated in Figs. 3 and 4, the phenological stages of sunflowers overlap with those of maize, zucchini, and soybean to some extent, potentially increasing the complexity of sunflower classification.

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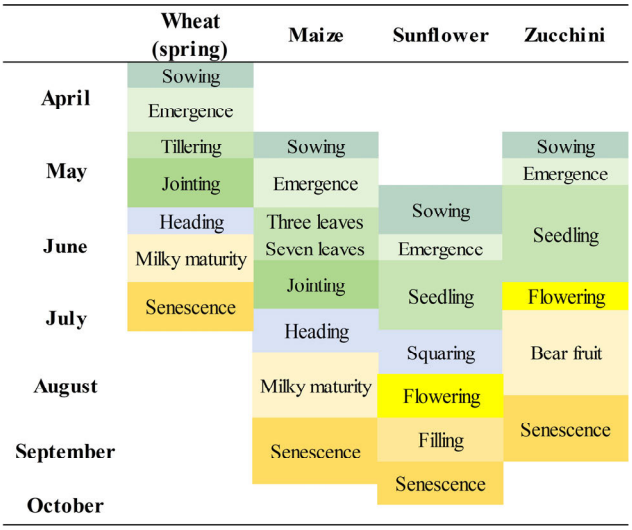


Fig. 3. Phenology calendar of the four major crops in S1.

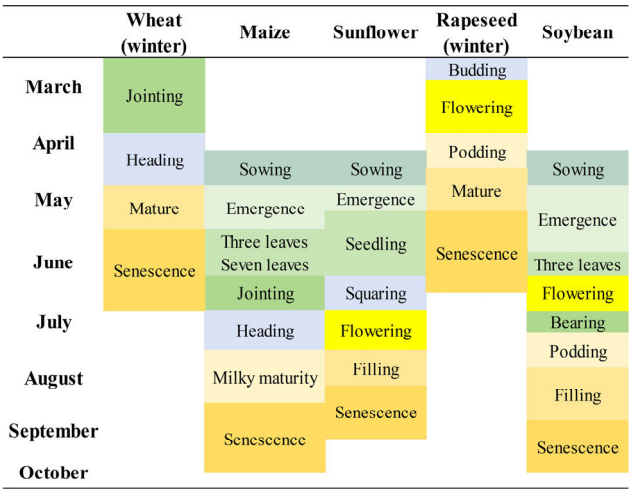


Fig. 4. Phenology calendar of the major crops in S2 and S3.

B. Analysis of Spectral Dynamics and Index Construction

1) Band Selection: To reveal the spectral differences between sunflower and other crops, this study used S1 training sample points to plot the spectral reflectance time-series curves of four crops across ten bands from May to September 2023 (Fig. 5). As shown in Fig. 5, the reflectance of sunflower [Fig. 5(a)] and maize [Fig. 5(b)] began to increase in early June, reached their peaks in early August, and then gradually declined. During the period from early August to early September, the spectral reflectance differences between the two crops were pronounced; this period, indicated by the two red dashed lines in the figure, can be considered the key time window for distinguishing sunflower from maize. In contrast, zucchini [Fig. 5(c)] and spring wheat [Fig. 5(d)] also exhibited significant reflectance differences from sunflower during the same period. Therefore, early August to early September was identified as the key time window for distinguishing sunflower from other crops [i.e., the classification period (CP)], corresponding to the late squaring to early filling stages of sunflower. To further select sensitive bands for constructing the SII, this study introduced the mean spectral difference index D_i to quantitatively evaluate each band's ability to distinguish sunflower from other crops during the CP. The calculation formula is as follows:

$$D_i = \frac{\left|\bar{\rho}_{sunflower,i} - \bar{\rho}_{other,i}\right|}{\bar{\rho}_{sunflower,i} + \bar{\rho}_{other,i}} \tag{1}$$

where $\bar{\rho}_{sunflower,i}$ and $\bar{\rho}_{other,i}$ represent the mean reflectance of sunflower and other crops, respectively, in the ith band.

Based on the D_i values of sunflower and other crops across different bands during the CP (Table III), bands B6 and B12 show complementary characteristics in distinguishing sunflower from other crops among the ten bands. Specifically, when distinguishing sunflower from maize, B6 exhibits the highest D_i value (0.1810), while B12 shows the lowest (0.0311). When distinguishing sunflower from zucchini, B12 has the highest D_i value (0.1965), whereas B6 has the lowest (0.1013). For distinguishing sunflower from spring wheat, B12 also demonstrates the strongest separability with a D_i value of 0.2610. Moreover, B6 is located in the red-edge region, which is highly sensitive to changes in vegetation chlorophyll content [47], [48], [49], while B12 lies in the SWIR region, which is sensitive to variations in canopy water content and exhibits reflectance that is negatively correlated with vegetation water content [50], [51]. The combined use of these two bands can comprehensively capture the physiological differences among crops during critical growth stages, thereby improving the stability and accuracy of sunflower identification. Therefore, this study selected B6 and B12 as the sensitive bands for constructing the SII.

2) SII: To further analyze the temporal characteristics of the sensitive bands, this study selected bands B6 and B12 to plot the spectral reflectance variation curves of four crops from May to September 2023 (Fig. 6). In the B6 band [Fig. 6(a)], the reflectance of sunflower began to increase in early June, reached its peak in early August, and then gradually declined. During the period from early August to early September (CP, indicated by the gray-shaded area), the reflectance of sunflower was significantly higher than that of the other crops (maize, spring wheat, and zucchini). Maize—the dominant crop in S1—exhibited relatively low and stable reflectance, resulting in a pronounced difference between the two. This indicates that the B6 band's sensitivity to chlorophyll variation can effectively distinguish sunflower from maize. In the B12 band [Fig. 6(b)], the reflectance of sunflower remained low and relatively stable during the CP, indicating that canopy water content was high and the crop was in a vigorous growth stage. Maize showed similar characteristics. In contrast, the reflectance of spring wheat and zucchini was significantly higher than that of sunflower during this period, demonstrating clear separability.

Overall, the differences in vegetation chlorophyll reflected by B6 and the differences in canopy water content reflected by B12 exhibit clear complementarity during the CP, indicating that sunflower maintains both high chlorophyll content and high canopy water content at this stage (characterized by lower SWIR2 values). Therefore, by combining the RE2 and SWIR2 bands, an SII was developed to effectively distinguish

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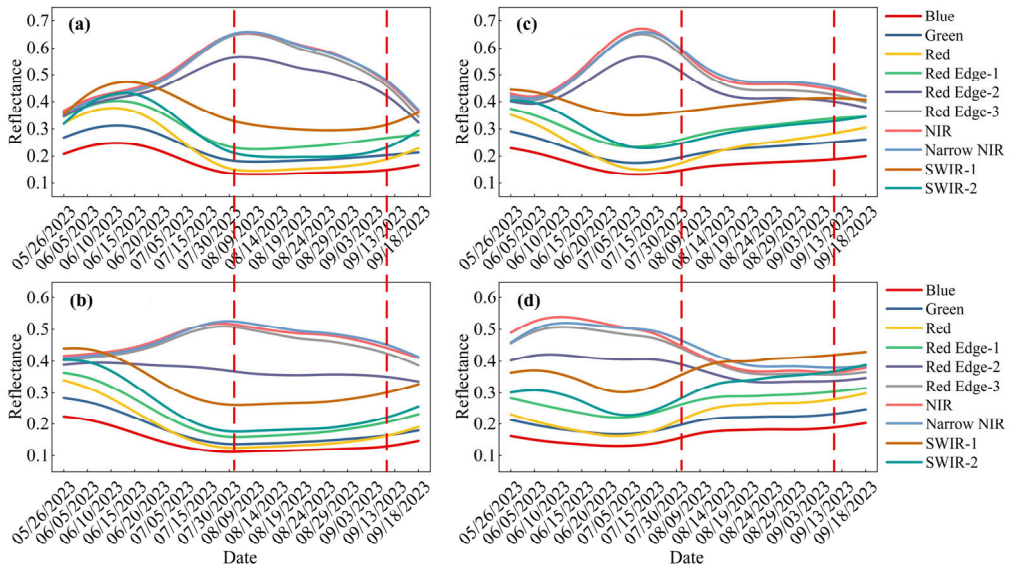


Fig. 5. Time-series curves of spectral reflectance across ten bands for four crops from S1 (2023). (a) Sunflower. (b) Maize. (c) Zucchini. (d) Spring wheat.

TABLE III
AVERAGE SPECTRAL SEPARABILITY (D_i) VALUES OF SUNFLOWER AND OTHER CROPS ACROSS DIFFERENT BANDS DURING THE CP

Band	Sunflower-Maize	Sunflower-Zucchini	Sunflower-Spring wheat
B2	0.0763	0.1144	0.1310
B3	0.1221	0.1046	0.0846
B4	0.0692	0.1952	0.2482
B5	0.1563	0.1271	0.0992
B6	**0.1810**	**0.1013**	0.1988
B7	0.1100	0.1345	0.2213
B8	0.1041	0.1130	0.2221
B8A	0.0954	0.1039	0.2188
B11	0.0500	0.1348	0.1441
B12	**0.0311**	**0.1965**	**0.2610**

sunflower from other crops. The calculation formula for the SII is shown as follows:

$$SII = \frac{\rho_{RE2} - \rho_{SWIR2}}{\rho_{RE2} + \rho_{SWIR2}} \tag{2}$$

where ρ_{RE2} and ρ_{SWIR2} represent the reflectance values of Sentinel-2 bands B6 and B12, respectively.

Fig. 7 illustrates the time-series variation of SII for the four major crops in S1 from May to September 2023. During the CP, the SII notably enhanced the sunflower signal, while the signals of the other three crops remained lower. This differentiation highlights the effectiveness of SII in distinguishing sunflower from other crops. Consequently, imagery acquired during the CP can be effectively utilized for sunflower identification and mapping.

C. NDVI and GNDVI

NDVI and GNDVI are widely used VIs in crop classification studies [40], [52], [53], [54] and have also demonstrated significant application value in sunflower classification [28], [39], [40], [55]. Therefore, in this study, NDVI and GNDVI were selected as benchmark indices to evaluate the performance of SII in sunflower identification. NDVI demonstrates a high degree of sensitivity to vegetation health and growth stages and is calculated using the red (ρ_{RED}) and NIR (ρ_{NIR}) bands of Sentinel-2 imagery [see (3)]. In contrast, GNDVI demonstrates a high degree of sensitivity to chlorophyll variations in vegetation and can also be utilized to assess plant water stress. It is derived from the green (ρ_{GREEN}) and NIR (ρ_{NIR}) bands [see (4)]. The following formulas represent the calculation of both indices:

$$NDVI = \frac{\rho_{NIR} - \rho_{RED}}{\rho_{NIR} + \rho_{RED}} \tag{3}$$

$$GNDVI = \frac{\rho_{NIR} - \rho_{GREEN}}{\rho_{NIR} + \rho_{GREEN}} \tag{4}$$

where ρ_{NIR} represents the reflectance of Sentinel-2's B8 band, while ρ_{GREEN} and ρ_{RED} correspond to the reflectance of the B3 and B4 bands, respectively.

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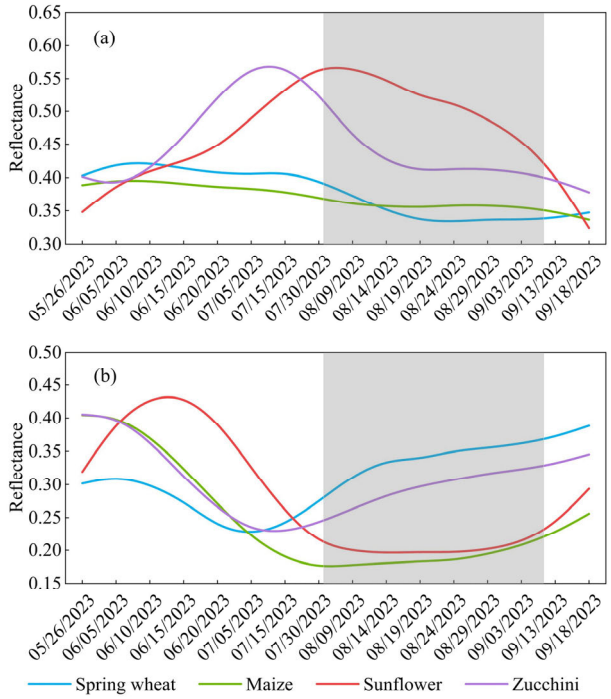


Fig. 6. Time-series variation of spectral reflectance for the four main crops in Sentinel-2 imagery at bands (a) B6 and (b) B12.

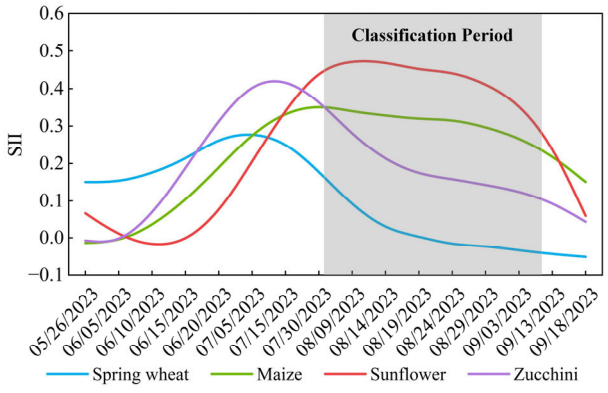


Fig. 7. Time-series variation of SII for the four major crops.

The utilization of (3) and (4), in conjunction with Sentinel-2 time-series data, resulted in the generation of NDVI [Fig. 8(a)] and GNDVI [Fig. 8(b)] time-series variation curves for the four primary crops in S1. As demonstrated in the figures, the optimal time window for distinguishing sunflowers from spring wheat and zucchini using NDVI and GNDVI is analogous to that of SII, primarily from early August to early September (corresponding to the late squaring to early filling stages). Conversely, the time window for distinguishing sunflowers from maize exhibits a substantial difference from that of SII, occurring primarily from early June to mid-July (sowing to seedling stages) or mid-September (late filling to senescence stages). This discrepancy reflects the fundamental differences in the construction principles of SII compared to NDVI and GNDVI.

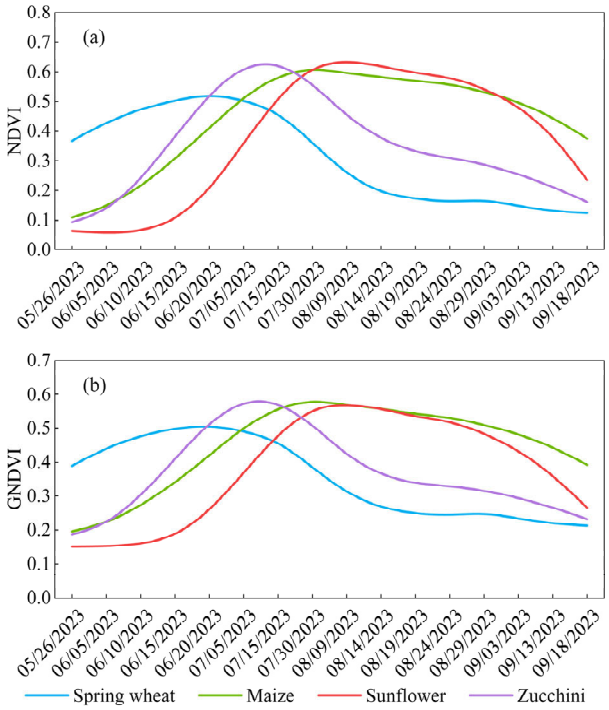


Fig. 8. Time-series variations of (a) NDVI and (b) GNDVI for the four major crops.

D. Relative Radiometric Correction

To eliminate radiometric inconsistencies among Sentinel-2 images acquired in different years—caused by variations in atmospheric conditions, solar elevation angles, and observation geometry—a relative radiometric correction was applied prior to assessing the temporal transferability of the SII. The correction procedure was implemented in Python using the rasterio and scikit-learn libraries. Specifically, one image was selected as the reference, and the top 5% of pixels exhibiting the smallest brightness variations were identified as pseudo-invariant features (PIFs). Linear regression models were then constructed for each spectral band to achieve radiometric matching across images. This method effectively reduces interannual radiometric differences and enhances spectral consistency, thereby providing a stable and reliable data foundation for cross-year sunflower identification using the SII. In S1, the images from August 14 to 24, 2023, were used to correct the images from August 19 to 29, 2021, respectively; in S3, the images from July 14, 2022, to August 8, 2022, were used to correct the images from July 9, 2020, to August 8, 2020, respectively.

E. ML Methods and Accuracy Assessment

To select the optimal index and evaluate its spatiotemporal transferability, the DT algorithm was used as the baseline classifier. To further assess the robustness of the optimal index, it was also incorporated as an input feature in RF and SVM models. This approach simplifies the identification process while verifying the index's applicability for automated sunflower mapping. DT is a supervised learning algorithm based on a

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tree structure that segments remote sensing image pixels into different categories according to predefined thresholds [56]. This approach offers several advantages, including its simple classification rules and high computational efficiency [57]. RF is an ensemble learning method that performs classification or regression by constructing multiple DTs and combining their predictive results. Due to its robustness and efficiency, RF has been widely applied in sunflower mapping [21], [55]. SVMs are nonparametric classifiers that maximize the margin between classes by identifying the optimal hyperplane. They address the issue of linear inseparability through kernel functions. It demonstrates robust generalization capability, particularly for high-dimensional data [58], [59], making it a prevalent choice in crop classification studies [15], [60]. Regarding parameter settings, following previous studies [61], the number of trees in the RF was set to 500. For the SVM, the kernel type and penalty parameter were set to radial basis function (RBF) and 50, respectively. The method for determining the threshold of the DT algorithm is described in Section III-F.

In this study, the producer's accuracy (PA), user's accuracy (UA), OA, and Kappa were calculated using the confusion matrix to assess classification accuracy at the pixel level [56]. The calculation formulas for these metrics are as follows:

$$PA = \frac{X_{ii}}{X_{+i}} \times 100\% \tag{5}$$

$$UA = \frac{X_{\mathrm{ii}}}{X_{i+}} \times 100\% \tag{6}$$

$$OA = \frac{\sum_{i=1}^{n} X_{ii}}{N} \times 100\% \tag{7}$$

$$Kappa = \frac{N\sum_{i=1}^{n} X_{ii} - \sum_{i=1}^{n} X_{i+}X_{+i}}{N^2 - \sum_{i=1}^{n} X_{i+}X_{+i}} \tag{8}$$

where X_{ii} represents the number of correctly classified pixels (i.e., the values on the diagonal of the confusion matrix); X_{i+} and X_{+i} denote the sums of the ith row and ith column, respectively; n is the number of classification categories; and N is the total number of pixels in the confusion matrix.

F. Classification Thresholds and DT

During the determination of classification thresholds, the following procedure was applied. First, based on the training samples, the reflectance values of each index for different crops within their respective classification time window were extracted, and their mean values were calculated. Next, the mean differences between sunflower and other crops were computed to identify the time points with the most pronounced spectral separability, which were then designated as the optimal distinguishing dates. Finally, the optimal thresholds were determined using a grid search approach. Following this procedure, the optimal image dates and corresponding thresholds for each index used to distinguish sunflower from other crops in the three study areas (S1–S3) were obtained (Table IV). Due to the diversity of crop types within the study areas, two thresholds (θ_1 and θ_2) were set for each index to exclude different interfering crops and achieve accurate sunflower identification. Due to variations in climatic conditions, crop types, and cultivation practices, the combination of SII thresholds may differ among study areas and therefore requires adjustment according to local conditions in practical applications. When assessing the temporal transferability of the SII, the thresholds in S1 showed slight interannual variations, primarily due to differences in annual climatic conditions, crop growth stages, and image acquisition parameters. It should be noted that the optimal thresholds for S1 (2021) were obtained through manual labeling of sample points on remote sensing images based on expert knowledge and visual interpretation, combined with the grid search method (Table IV). Specifically, 130 sample points were annotated for each of sunflower and maize, while 50 and 30 points were annotated for zucchini and wheat, respectively. In contrast, the thresholds for S3 remained consistent across different years, further demonstrating the good interannual stability and adaptability of the SII. During the construction of the DT, an intersection logic (AND) was applied to further improve classification accuracy. Specifically, a pixel was classified as sunflower only if it simultaneously met both thresholds θ_1 and θ_2 (for example, index $\geq \theta_1$ and index $\leq \theta_2$); otherwise, it was classified as nonsunflower.

IV. Results

A. Index Screening and Accuracy Validation

Using the DT algorithm together with the optimal thresholds of NDVI, GNDVI, and SII (Table IV), the sunflower distribution in S1 for 2023 was extracted. The sunflower classification results generated by the three indices are shown in Fig. 9(a)–(c). The results indicate that the sunflower distribution identified using NDVI [Fig. 9(a)] and GNDVI [Fig. 9(b)] is significantly larger than that identified using SII [Fig. 9(c)]. The disparities are predominantly concentrated in the eastern and western regions of S1, while the central region exhibits comparatively minor variations.

To enhance the visual comparison, three 5 × 5 km plots within S1 were selected to demonstrate the differences in classification results among the three indices (see Fig. 10). To ensure a fair comparison, no postclassification processing was applied to the sunflower classification results derived from the three indices. The classification outcomes of NDVI and GNDVI manifest conspicuous omissions and misclassifications, impeding the acquisition of accurate and complete sunflower plots in complex cropping patterns, as indicated by the red circles in Fig. 10. In contrast, SII demonstrates a superior capacity for sunflower field identification, exhibiting minimal misclassification or omission, as evidenced by the blue circles in Fig. 10. In Fig. 10(a) and (b), NDVI and GNDVI misclassified substantial areas of maize as sunflower and exhibited notable omission of sunflower regions. Conversely, SII demonstrated a marked aptitude for differentiating between the sunflower and maize, thereby furnishing comprehensive sunflower plot data. In Fig. 10(c), NDVI and GNDVI were unable to accurately extract sunflower in areas with mixed cropping and highly fragmented farmland, whereas SII effectively identified sunflower with minimal omission. In summary, SII has achieved satisfactory results, enabling more accurate differentiation between sunflower and other crops, with less noise and well-defined boundaries.

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TABLE IV
OPTIMAL IMAGE DATES AND CORRESPONDING THRESHOLDS OF VARIOUS INDICES FOR DISTINGUISHING SUNFLOWER FROM OTHER CROPS

Site	Index	Time phase -1	θ_1	Excluding crop -1	Time phase -2	θ_2	Excluding crop -2
S1	NDVI	20230620	LE 0.27	Maize	20230814	GE 0.51	Zucchini, Spring wheat
	GNDVI	20230620	LE 0.30	Maize	20230814	GE 0.48	Zucchini, Spring wheat
	SII	20230814 20210819	GE 0.39 GE 0.32	Maize, Spring wheat	20230824 20210829	GE 0.24 GE 0.16	Zucchini
S2	SII	20220703	GE 0.15	Maize, Soybean	20220721	GE 0.09	Winter wheat, Winter rapeseed, Other crops
S3	SII	20220714 20200709	GE 0.11	Winter wheat, Winter rapeseed, Other crops	20220808 20200808	LE 0.22	Maize, Soybean

Note: LE and GE denote 'less than or equal to (≤)' and 'greater than or equal to (≥)', respectively.

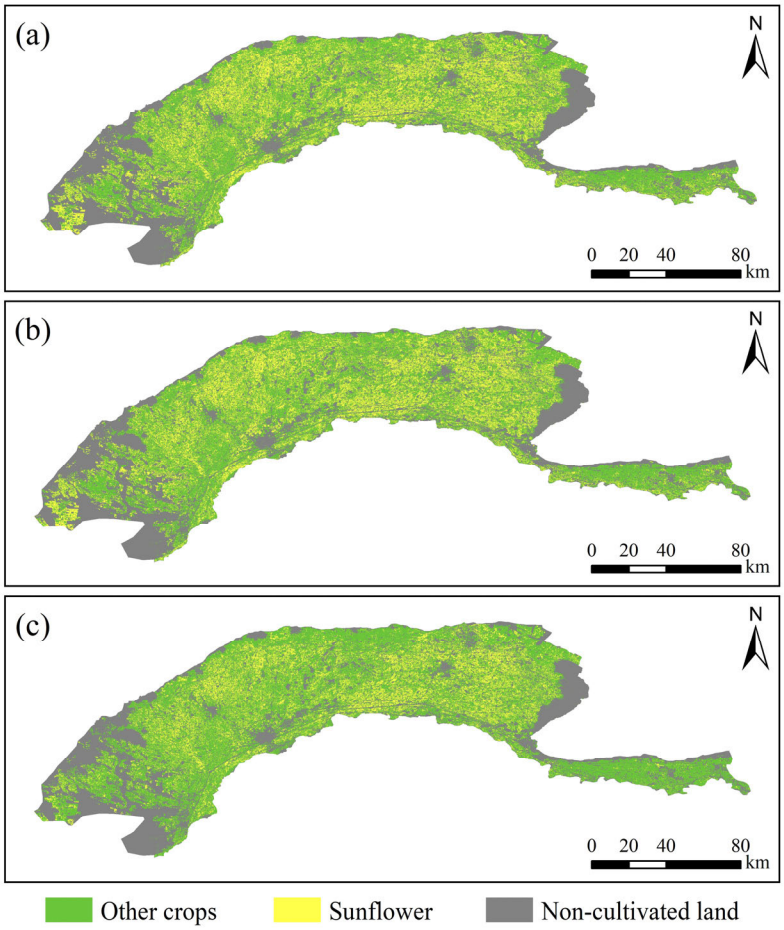


Fig. 9. Sunflower classification results in S1 in 2023 using three VIs: (a) NDVI; (b) GNDVI; and (c) SII.

The classification accuracy of NDVI, GNDVI, and SII for sunflower identification was evaluated (Table V) using the validation data from S1 and the four metrics mentioned in Section III-E. The findings suggest that SII attained the highest level of accuracy, with an OA of 94.39%, a substantial improvement over NDVI (OA = 81.31%) and GNDVI (OA = 78.51%). Meanwhile, SII exhibited the highest Kappa coefficient (0.89), which was 0.26 and 0.32 higher than those of

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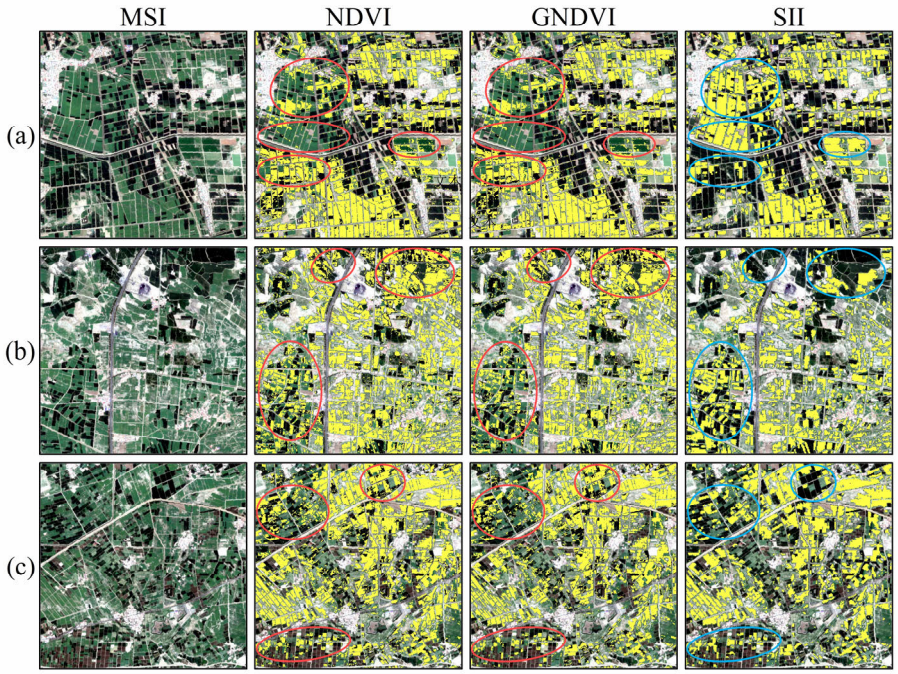


Fig. 10. Comparison of classification results for three indices in three sample plots in S1 in 2023: (a) plot 1; (b) plot 2; (c) plot 3. From left to right in each plot: multispectral image, NDVI, GNDVI, and SII classification results. Each plot measures 5 × 5 km.

TABLE V
COMPARISON OF CLASSIFICATION ACCURACY FOR VARIOUS INDICES IN S1

Index	OA (%)	Kappa	PA (%)		UA (%)	
			Sunflower	Non sunflower	Sunflower	Non sunflower
NDVI	81.31	0.63	83.33	79.25	80.36	82.35
GNDVI	78.51	0.57	75.93	81.13	80.39	76.79
SII	94.39	0.89	92.59	96.23	96.15	92.73

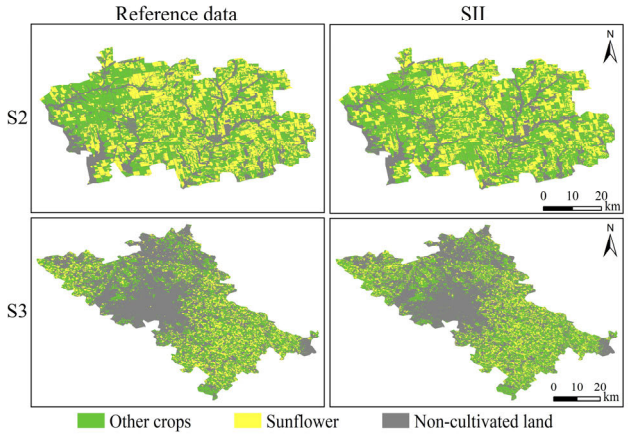


Fig. 11. Sunflower classification results for S2 and S3 in 2022 based on SII. Reference data are sourced from EUCROPMAP 2022.

NDVI and GNDVI, respectively. Moreover, both the PA and UA for sunflower and nonsunflower classes surpassed 92% in SII, underscoring their capacity for high-accuracy sunflower identification in complex agricultural landscapes. In summary, SII proved to be the optimal index for sunflower classification.

B. Analysis of the Transferability of SII

1) Spatial Transferability of SII: To evaluate the applicability of SII across different regions, the sunflower distribution in S2 and S3 in 2022 was extracted using the DT algorithm and the optimal thresholds (Table IV). As demonstrated in Fig. 11, the sunflower classification results generated by SII exhibited a high degree of consistency with the reference data from the study areas, despite varying climatic conditions. This consistency underscores the efficacy and robustness of SII in distinguishing sunflowers from various other crop types. In S2, sunflower fields manifest a relatively uniform spatial distribution, characterized by larger and more regularly shaped parcels. In contrast, in S3, sunflower fields are more sparsely distributed, exhibiting fragmented and irregularly shaped plots (Fig. 11). Notably, SII successfully captures sunflower fields with different distribution patterns and shapes, highlighting its adaptability across diverse agricultural landscapes.

To more intuitively illustrate the classification performance of the SII in different regions, representative areas in S2 and S3 were selected for enlarged comparison (Fig. 12). In S2, the classification results of SII are largely consistent with the reference data. However, some misclassifications in the

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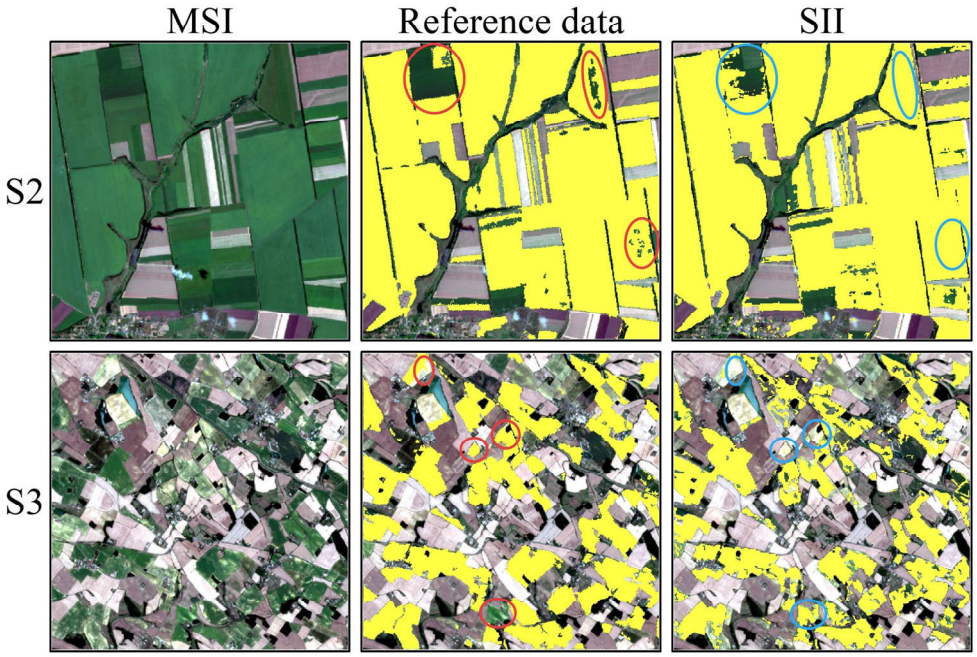


Fig. 12. Comparison of reference data and SII classification results for sample plots in S2 and S3 in 2022 (from left to right: multispectral imagery, reference data, and SII classification results). Each plot has a size of 5 × 5 km. The reference data are sourced from EUCROPMAP 2022.

TABLE VI
CLASSIFICATION ACCURACY BASED ON SII FOR S2 AND S3 IN 2022

Site	OA (%)	Kappa	PA (%)		UA (%)	
			Sunflower	Non sunflower	Sunflower	Non sunflower
S2	92.41	0.84	89.66	94	89.66	94
S3	93.10	0.83	91.30	93.75	84	96.77

SII results (indicated by blue circles) and omissions in the reference data (highlighted by red circles) are observed. In S3, SII demonstrated a high degree of accuracy in sunflower classification. Nevertheless, the reference data exhibited tendencies to misclassify bare land or fallow fields (i.e., farmland after the harvest of winter crops) as sunflowers and demonstrated some omission errors (highlighted by red circles). Research conducted in France has indicated that sunflowers in this region are often sparsely distributed at very low field densities or intercropped with other crops. This phenomenon poses significant challenges for accurate sunflower identification [21].

With regard to the validation of accuracy, SII demonstrated consistent performance in both S2 and S3, as evidenced in Table VI. The OA for S2 and S3 was 92.41% and 93.10%, respectively, with Kappa coefficients of 0.84 and 0.83. Additionally, the PA and UA of sunflowers substantiate the efficacy of SII in accurately identifying sunflowers, while effectively mitigating the impact of nonsunflower crops on the classification outcomes. These results indicate that SII has strong spatial transferability and can be applied to sunflower identification tasks across multiple regions.

2) Temporal Transferability of the SII: To evaluate the stability of SII for sunflower identification across different years within the same region, a cross-year classification was conducted for S1 in 2021 using the optimal thresholds (Table IV). The resulting spatial distribution of sunflowers in S1 in 2021 [Fig. 13(a)] shows strong agreement with the classification results for the Hetao Plain reported in other studies for the same year [28], [33], thereby further confirming the reliability of the SII for sunflower identification. In addition, three representative plots of 5 × 5 km were selected for enlarged visualization, corresponding to contiguous sunflower fields [Fig. 13(b1) and (b2)], sunflower–maize intercropping zones [Fig. 13(c1) and (c2)], and mixed cropping areas with multiple crop types [Fig. 13(d1) and (d2)]. Across all planting patterns, the SII demonstrated high classification accuracy, with very limited misclassification and only slight omission occurring near the edges of low-density sunflower fields. A cross-year transfer experiment was also performed for S3 using the same thresholds (Table IV). As shown in Fig. 14, the SII-derived sunflower distribution for 2020 [Fig. 14(b)] aligns closely with the reference data [Fig. 14(a)], indicating strong temporal transferability of the SII. Enlarged comparisons for two representative plots [Fig. 14(c) and (f)] reveal that the SII classification results [Fig. 14(e) and (h)] maintain high consistency with the reference datasets [Fig. 14(d) and (g)], with only minor omissions observed in a few localized areas. Notably, both the Theia Land Cover Map 2020 and EUCROPMAP 2022

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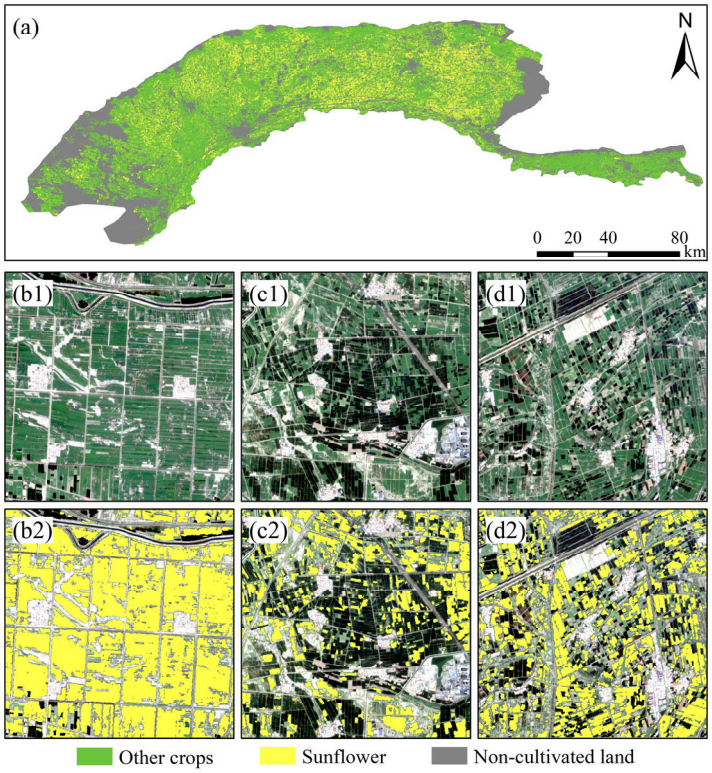


Fig. 13. Sunflower classification results for S1 in 2021. (a) Overall classification results based on the SII. (b1), (c1), and (d1) Sentinel-2 imagery of the three selected 5 × 5 km plots. (b2), (c2), and (d2) Enlarged SII classification results corresponding to the selected plots.

tend to misclassify bare soil as sunflower, whereas the SII effectively avoids such errors.

C. Area Estimation

This study evaluated the accuracy of the classified areas by calculating the percentage error between the predicted sunflower areas and the reference areas (Table VII). The predicted areas were derived from the SII classification results, while the reference areas were obtained from multiple sources: for S1, official statistical data from Bayannur for 2021 and 2023 were used; for S2, EUCROPMAP 2022 was used; and for S3, both the Theia Land Cover Map 2020 and EUCROPMAP 2022 were used. For S1, the sunflower planting areas in Wuyuan County, Linhe District, and Hangjinhou Banner were selected for comparison. The results show that the predicted area based on 2023 imagery was approximately 7.35% lower than the statistical data, indicating high reliability of the SII for that year. In contrast, the 2021 classification results underestimated the area by 19.37%, which may be attributable to several factors, including the fragmentation of arable land, low planting density in some sunflower plots leading to identification difficulties, differences in image acquisition timing, crop phenology variations, and interannual spectral changes. In S2, the sunflower area estimated by the SII was 12.94% lower than the reference area. This discrepancy may be due to the inadvertent exclusion of sunflower fields during the image-clipping process using farmland boundary data, as well as a minor degree of omission in the classification process. For S3, the predicted sunflower area for 2020 was 2.20% higher than the reference data, whereas the predicted area for 2022 was 1.09% lower. These discrepancies are primarily attributable to the generally low planting density of sunflowers in France and the frequent interspersed distribution of sunflowers and bare soil within fields, which can interfere with the accurate identification of sunflower plots. In addition, misclassification of bare or fallow land as sunflower in EUCROPMAP 2022 may have contributed to the overestimation of area.

D. Robustness Evaluation of SII Across Different Classifiers

As demonstrated in Sections IV-A and IV-B, SII achieved satisfactory accuracy in the three study areas (S1–S3). To further assess the robustness of this index, SII was applied to the RF and SVM classifiers on the Google Earth Engine (GEE) platform for binary classification of sunflower and nonsunflower areas. To ensure the reliability of the classification results, mean Sentinel-2 images within the respective CPs were selected as the data source for all three study areas (S1–S3). Considering that different crops exhibit pronounced spectral differences within the CP and that these characteristics vary notably over time, for instance, in S1 (2023), the SII of sunflower was substantially higher than that of other crops in August, while at the beginning of September, its SII declined much more rapidly than that of other crops, particularly maize (Fig. 7). This pattern indicates that sunflower reaches the maturity stage earlier than other crops, providing a key basis for its identification. This study, therefore, leveraged these temporal spectral variations to enhance the discriminative power of the classifier and to streamline the classification

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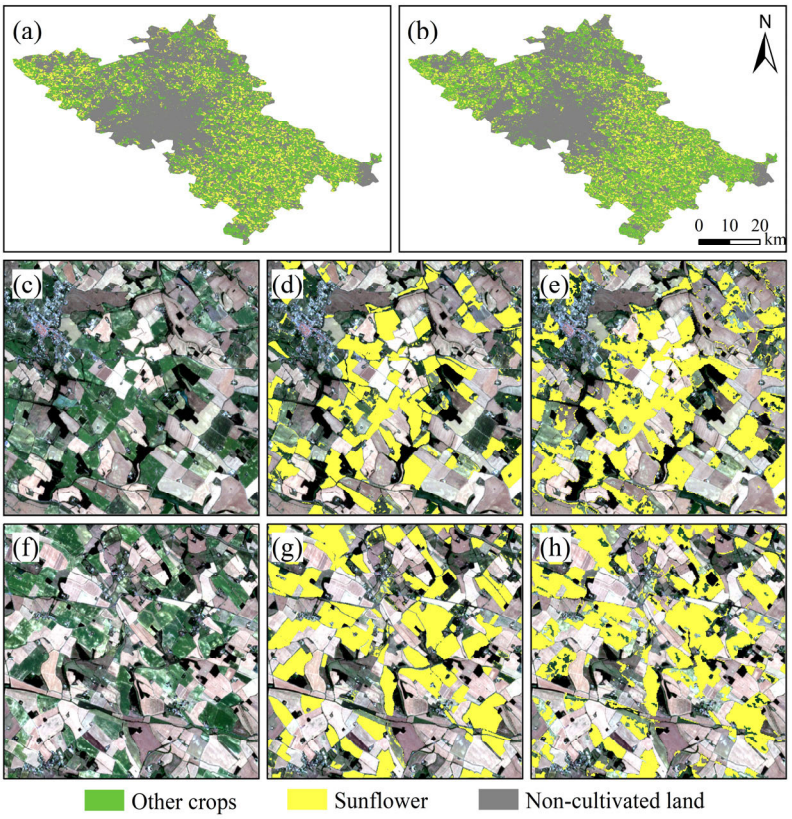


Fig. 14. Comparison of the reference data and SII classification results for S3 in 2020. (a) Reference data: Theia Land Cover Map 2020. (b) Overall SII classification results. (c)–(h) Enlarged views of two selected 5 × 5 km plots showing Sentinel-2 imagery, reference data, and SII classification results, respectively.

TABLE VII
COMPARISON OF PREDICTED AND REFERENCE SUNFLOWER AREAS (KHA)

Site	Predicted area (kha)	Reference area (kha)	Percentage error (%)	Year
S1	227.76★	-	-	2023
	152.92*	165.05*	-7.35*	
	212.95★	-	-	2021
	140.17*	173.84*	-19.37*	
S2	61.71	70.88	-12.94	2022
S3	36.36	36.76	-1.09	2022
	39.54	38.69	2.20	2020

Note: '★' indicates the sunflower classification results in the study area. '*' indicates sunflower data from Wuyuan County, Linhe District, and Hangjinhou Banner.

process. Monthly intervals were used as the temporal resolution for the input features. Specifically, for S1, SII images from August 2023 (August 1–31) and early September 2023 (September 1–10) were used as input features; for S2 and S3, SII images from July 2022 (July 1–31) and early August 2022 (August 1–10) were selected. A comparative analysis (Table VIII) revealed that both the RF and SVM classifiers achieved high classification accuracy across the three study areas (OA > 90% and Kappa ≥ 0.81). Among them, SVM outperformed RF overall, particularly in S2 and S3. With regard to the performance of RF and SVM in terms of PA and UA for sunflower and nonsunflower classification, both classifiers exhibited strong performance across all study areas. Furthermore, the integration of SII with the DT algorithm yielded notable PA and UA values, as presented in Tables V and VI. These results suggest that SII can be effectively integrated with various ML classifiers to accurately identify sunflowers in regions with high landscape complexity, further demonstrating its robustness.

V. DISCUSSION

A. Advantages of SII

The SII proposed in this study enables accurate sunflower identification under diverse environmental conditions using Sentinel-2 data, providing an efficient and reliable tool for remote sensing applications in sunflower research. In practical applications, SII demonstrates significant advantages, particularly in the following three aspects.

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TABLE VIII
CLASSIFICATION ACCURACY OF SII USING RF AND SVM CLASSIFIERS

Site	Methods	OA (%)	Kappa	PA (%)		UA (%)	
				Sunflower	Non sunflower	Sunflower	Non sunflower
S1	RF	90.65	0.81	85.19	96.23	95.83	86.44
	SVM	90.65	0.81	87.04	94.34	94	87.72
S2	RF	92.41	0.84	89.66	94	89.66	94
	SVM	93.67	0.86	86.21	98	96.15	92.45
S3	RF	93.1	0.82	82.61	96.88	90.48	93.94
	SVM	96.55	0.91	95.65	96.88	91.67	98.41

1) SII is constructed based on differences in spectral sensitivity to vegetation physiological characteristics across different bands, allowing for effective differentiation between sunflowers and other crops. Studies have shown that the RE2 band is highly sensitive to variations in chlorophyll content in vegetation [47], [49], [62]. During specific periods of the crop growth cycle, crops with higher chlorophyll content generally exhibit significantly higher reflectance in the RE2 band than those with lower chlorophyll levels. For instance, throughout most of the growing season, sunflowers exhibit significantly higher RE2 reflectance compared to maize and spring wheat. Similarly, when compared to zucchini, sunflowers display higher RE2 reflectance from early August to early September, corresponding to the CP. This highlights the RE2 band's crucial role in distinguishing sunflowers from other crops. Unlike RE2, the SWIR2 band is located in the water absorption spectrum region, allowing it to accurately capture differences in vegetation canopy water content [51], [63], [64]. Generally, the higher the vegetation canopy water content, the lower the reflectance in SWIR2 [50]. Additionally, vegetation with denser canopies typically has higher canopy water content [63], [65]. From a physiological and morphological perspective, sunflowers are characterized by tall plants, dense canopies, and a well-developed root system [66]. Their high transpiration rate results in greater water retention in the leaves, leading to lower SWIR2 reflectance compared to other crops, especially during the CP. Maize exhibits similar characteristics [67], but in the RE2 band, there is a significant difference in reflectance between maize and sunflowers, providing a key criterion for distinguishing between the two crops. Conversely, compared to sunflowers, spring wheat and zucchini have shorter plants, thinner canopies, more dispersed leaves, and shallower root systems, leading to lower transpiration rates and lower water content in the leaves. Consequently, these crops exhibit higher SWIR2 reflectance, particularly during the CP. Among the analyzed crops, sunflowers are the only ones that simultaneously exhibit high RE2 reflectance and low SWIR2 reflectance during the CP. In summary, leveraging the spectral characteristics of the RE2 (B6) and SWIR2 (B12) bands enables the development of an index capable of accurately identifying sunflowers.
2) Compared to traditional VIs such as NDVI and GNDVI, SII offers a longer time span and a more suitable classification window. Taking S1 as an example, SII requires only images from the late squaring to early filling stages of sunflowers (early August to early September) to accurately distinguish sunflowers from other crops. During this period, crops reach their growth peak, exhibiting distinct spectral differences, which enhance classification accuracy. Additionally, the classification window provided by SII is relatively concentrated, minimizing unnecessary data processing and improving classification efficiency. In contrast, NDVI and GNDVI require multiple time windows for effective classification. For example, distinguishing between sunflowers and maize requires images from either the sunflower sowing to seedling stage (early June to mid-July) or the late filling to senescence stage (mid-September). However, during these phases, crops are either in early growth or senescence, when their spectral features are either not well-developed or have significantly weakened, making accurate classification more challenging. When distinguishing sunflowers from zucchini and spring wheat, the time window required by NDVI and GNDVI is similar to that of SII. Nevertheless, the classification windows for NDVI and GNDVI are more dispersed, requiring the processing of a larger volume of remote sensing data, thereby increasing computational workload. In summary, SII offers a well-defined classification window with greater efficiency, making it particularly suitable for sunflower remote sensing identification.
3) SII demonstrates excellent applicability, stability, and robustness. The results indicate that, when combined with the DT classifier, SII achieved high-accuracy sunflower identification in S1 for 2023, successfully extracting complete sunflower parcels even under complex planting conditions. When applied to S2 and S3 for 2022, SII likewise produced high-accuracy classification results, further demonstrating its strong applicability. Moreover, when transferred across years to S1 in 2021 and S3 in 2020, SII was still able to reliably extract complete sunflower parcels and accurately estimate their

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areas, indicating that the index exhibits strong stability as well as robust temporal and spatial generalizability. Compared to complex classification models [21], [32], [33], SII offers a simpler computational process, requires fewer input variables, and achieves higher computational efficiency, making it ideal for resource-limited environments. Furthermore, when integrated with RF and SVM classifiers, SII continues to demonstrate strong performance, reinforcing its robustness. The ability of SII to perform well across different classification models suggests that it can adapt to various algorithmic requirements while reducing reliance on complex models. This flexibility makes SII particularly valuable for large-scale sunflower monitoring and mapping applications.

B. Limitations and Prospects

A comparison of the SII with traditional VIs reveals significant advantages in the recognition of sunflowers. However, it still has certain limitations. First, the use of optical remote sensing data for crop mapping is inherently affected by cloud cover, which remains an unavoidable challenge [68], [69]. Cloud contamination not only reduces the availability of satellite imagery but also undermines the reliability of reflectance data in the RE2 and SWIR2 bands, thereby adversely affecting the accuracy, timeliness, and completeness of sunflower classification using SII. In practical applications, this suggests that the applicability of SII may be constrained in regions with frequent cloud cover or during rainy seasons. It is worth noting, however, that SII can support sunflower mapping by utilizing imagery within an extended temporal window. In addition to using cloud-free median or mean composite images to mitigate cloud interference, future improvements may include multitemporal image compositing, cloud-filling algorithms based on time-series analysis [e.g., harmonic analysis and Spatial and Temporal Adaptive Reflectance Fusion Model (STARFM)], and the integration of optical and radar data to further enhance data continuity and classification stability. A second important consideration is the need to understand the phenological characteristics of sunflower within the target agricultural region—particularly for determining the CP—which is a prerequisite for applying SII. While this study obtained CP information for China, Ukraine, and France, extending the use of SII to other regions requires accurate knowledge of local phenological patterns; otherwise, classification biases may arise due to climatic differences or variations in cropping systems. Future research may achieve rapid and precise CP determination through automatic extraction based on remote-sensing time-series features or through model-based approaches that integrate meteorological and ground-observation data, thereby enhancing the transferability and classification efficiency of SII. Third, the development of VIs based on crop-specific characteristics and the achievement of rapid identification through threshold setting has become a major focus in crop extraction studies [57], [70]. However, in regions characterized by intricate planting structures, a single threshold may encounter difficulties in effectively differentiating between crops with similar spectral characteristics to the target crop, leading to a significant decline in classification accuracy [19]. In the classification process, combining SII with various ML algorithms can mitigate this issue to some extent, but it may also lead to a "salt-and-pepper effect." Recent studies have shown that DL algorithms capable of extracting complex features from imagery—such as recurrent neural networks (RNNs), convolutional neural networks (CNNs), and U-Net—are developing rapidly [36], [71]. Future research could explore integrating SII with DL methods to fully exploit the spatiotemporal characteristics of remote sensing imagery, thereby achieving higher classification accuracy. In addition, leveraging the complementary advantages of optical and radar data, combined with DL approaches, to enable large-scale, multisource, and multitemporal sunflower mapping will become an important direction for future developments in remote sensing-based crop identification.

VI. Conclusion

In this study, we developed the SII based on the RE2 (B6) and SWIR2 (B12) bands of Sentinel-2 imagery and validated its performance across three major sunflower-producing countries. The key findings are as follows.

1) SII provides an extended classification window, spanning from the late squaring to the early filling stages, with broad applicability, enabling accurate sunflower identification.
2) In S1 (2023), SII was identified as the optimal index, achieving significantly higher classification accuracy than NDVI and GNDVI. Its strong performance was maintained in both spatial transfer (S2 and S3 in 2022) and temporal transfer (S1 in 2021 and S3 in 2020) applications.
3) In the estimation of classified area, minor errors may arise from interannual differences in image acquisition timing and crop phenology, uncertainties in farmland boundary data, overestimation of sunflower planting area in the reference datasets, and potential omission errors during the classification process.
4) When integrated with RF and SVM, SII achieved satisfactory results across S1–S3 (OA $>$ 90% and Kappa $\geq$ 0.81).
5) From an application perspective, the large-scale implementation of SII requires comprehensive consideration of the impacts of cloud contamination and regional differences in sunflower phenology.

Further integration with advanced classification algorithms is expected to enhance its application potential. In summary, SII can be effectively combined with various ML classifiers to achieve accurate sunflower identification in complex agricultural environments, while maintaining good stability and robustness across different temporal and spatial scales, demonstrating its potential for large-scale sunflower mapping.

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